

Target Hamiltonian

$$f_q, \alpha$$

Physics targets

Physics relation

$$f_q \approx \sqrt{8f_J f_C} - f_C$$
$$\alpha \approx -f_C$$

Scale inputs

Min--max scaling

Training-set scalers

Machine Learning

Inverse MLP

TransmonCross model

$$f_q, \alpha \rightarrow \ell_{\text{claw}}, s_{\text{ground}}, \ell_{\text{cross}}$$

Refine if needed

Validation

Compare

Percent error

Validate or refine

Recovered
Hamiltonian

$$\hat{f}_q, \hat{\alpha}$$

Forward check

Quantum Metal layout

Surrogate or Ansys Q3D

Decode prediction

Physical units

Valid geometry range